
Introduction, Exponential Smoothing & Forecast Evaluation

ECON 8310: Business Forecasting — Lecture 1

Department of Economics
University of Nebraska at Omaha
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Lecture Outline

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Why Forecasting Matters

Every business decision made today depends on expectations about tomorrow.

Forecasting Is Everywhere in Business

Domain	The question someone is actually asking
Operations & supply chain	How much inventory should Walmart order for Black Friday? How many nurses does a hospital need next Tuesday night?
Finance & risk	Will credit card defaults rise next quarter? What will the Fed Funds rate be in six months?
Marketing & strategy	How many units will this new product sell in year one? What is the lifetime value of this customer cohort?
Macro & public policy	Will GDP grow by 2% or 3% next year? What will unemployment be in Q4?

Every one of these is a *decision* waiting on a number. **A decision is only as good as the forecast it rests on.**

The Basic Setup

Time Series

A **time series** is an ordered sequence of observations $\{y_1, y_2, \dots, y_T\}$ where t indexes time.

Notation used throughout the course:

- y_t — observed value at time t
- h — **forecast horizon**, how many periods ahead
- $\hat{y}_{t+h|t}$ — forecast of y_{t+h} made using information at time t
- $e_t = y_t - \hat{y}_{t|t-1}$ — **forecast error**

Forecasting monthly retail sales, $h = 1$ is next month and $h = 12$ is a year out. The problem gets harder as h grows, because uncertainty compounds.

Under squared-error loss the optimal forecast is the **conditional expectation**,

$\hat{y}_{t+h|t} = \mathbb{E}[y_{t+h} \mid \mathcal{F}_t]$, where $\mathcal{F}_t$ is everything known at time t .

Components of a Time Series

Any series can be decomposed into four parts (additive form): $y_t = T_t + S_t + C_t + I_t$.

- T_t **Trend** — long-run direction
- S_t **Seasonality** — regular, calendar-driven pattern
- C_t **Cycle** — medium-run business-cycle swings
- I_t **Irregular** — random shocks

Which parts matter depends entirely on the series: retail sales are trend + seasonality, quarterly GDP is trend + cycle, daily stock returns are mostly irregular, and energy demand has all four.

Ignoring seasonality causes large, systematic errors. A non-seasonal model on monthly retail data will consistently under-forecast December and over-forecast January — every single year.

Always Beat a Benchmark First

Before presenting any forecast model, check that it beats a **naïve benchmark**. A model that cannot beat the baseline has zero value.

Benchmark	Forecast	Use when
Naïve	$\hat{y}_{t+h t} = y_t$	Asset prices, random walks
Seasonal naïve	$\hat{y}_{t+h t} = y_{t+h-m}$	Any series with a season
Mean	$\hat{y}_{t+h t} = \bar{y}_t$	No trend, no season
Drift	$\hat{y}_{t+h t} = y_t + h\hat{c}$	Trend, no season

This is not a formality. In the M4 Competition many machine learning models *failed* to beat simple exponential smoothing (Makridakis et al. 2020).

Exponential Smoothing

Recent observations should carry more weight than old ones.

The Problem with Equal Weights

Most classical methods treat every historical observation equally — but what if the world **changes**? When a retailer's sales pattern shifts after a store renovation, equal-weighting the pre-renovation data buys months of systematic over- or under-forecasting.

Exponential weighting assigns geometrically declining weights, so recent observations matter more:

Observation	Age	Weight
y_T	0	α
y_{T-1}	1	$\alpha(1 - \alpha)$
y_{T-2}	2	$\alpha(1 - \alpha)^2$

$\alpha \in (0, 1]$ is the forgetting rate. High α = short memory; low α = long memory.

Simple Exponential Smoothing (SES)

SES Update Equation

Let ℓ_t be the **level** at time t :

$$\ell_t = \alpha y_t + (1 - \alpha) \ell_{t-1}, \quad 0 < \alpha \leq 1$$

Forecast: $\hat{y}_{T+h|T} = \ell_T$ for all horizons $h \geq 1$ (flat)

Large α (e.g. 0.8):

- Reacts quickly to shocks
- Noisy, volatile forecasts
- Good for rapidly changing series

Small α (e.g. 0.1):

- Slow to adapt
- Smooth, stable forecasts
- Good for slowly evolving series

SES nests both benchmarks: $\alpha = 1$ gives the naïve forecast, and $\alpha \rightarrow 0$ gives the historical mean. *Use when there is no trend and no seasonality.*

Holt's Method: Adding Trend

SES produces a *flat* forecast. If the series trends, track both the level and the slope.

Holt's Linear Exponential Smoothing (Holt 2004)

Level: $\ell_t = \alpha y_t + (1 - \alpha) (\ell_{t-1} + b_{t-1})$

Trend: $b_t = \beta^* (\ell_t - \ell_{t-1}) + (1 - \beta^*) b_{t-1}$

Forecast: $\hat{y}_{T+h|T} = \ell_T + h b_T$

Read ℓ_t as where the series *is* now and b_t as its current growth rate.

Linear extrapolation gets unrealistic at long horizons. For $h > 6$ use the **damped trend** variant, $\hat{y}_{T+h|T} = \ell_T + \sum_{j=1}^h \phi^j b_T$ with $0 < \phi < 1$. At $\phi = 0.85$ the forecast converges to a finite ceiling instead of growing forever.

Holt-Winters: Level, Trend, and Seasonality

Holt-Winters Additive Method (Winters 1960)

$$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1}$$

$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m}$$

Forecast: $\hat{y}_{T+h|T} = \ell_T + h b_T + s_{T+h-m}$ (m = seasonal period)

Additive seasonality:

- Seasonal swings are constant in size
- Same $\pm$ amount every December

Multiplicative seasonality:

- Swings grow with the level
- Retail sales, tourism

In US retail sales the December spike grows every year as overall sales grow, which makes it **multiplicative**. Requires at least $2m$ observations to initialize ($m = 12$ for monthly data).

The ETS Framework: A Unified View

Hyndman and Athanasopoulos (2021) unify every exponential smoothing method as a **state space model**: Error $\times$ Trend $\times$ Seasonal.

Component	Options	Meaning
Error (E)	A, M	Additive / Multiplicative
Trend (T)	N, A, A_d	None / Additive / Damped
Seasonal (S)	N, A, M	None / Additive / Multiplicative

That gives 15 valid combinations, and the ones we have just built are among them: **ETS(A,N,N)** is SES, **ETS(A,A,N)** is Holt linear, **ETS(A, A_d ,N)** is Holt damped, **ETS(A,A,A)** is Holt-Winters additive, and **ETS(A,A,M)** is Holt-Winters multiplicative.

Automatic selection: fit all 15 and pick the minimum AIC. In Python, `statsmodels.tsa.exponential_smoothing.ets.ETSModel`.

Forecast Evaluation

Forecasting without evaluation is not science — it is storytelling.

How Do We Measure Forecast Accuracy?

Given out-of-sample errors $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$:

Scale-dependent:

$$\text{RMSE} = \sqrt{\frac{1}{H} \sum_{h=1}^H e_{T+h}^2}$$

$$\text{MAE} = \frac{1}{H} \sum_{h=1}^H |e_{T+h}|$$

Same units as y_t . Compare models on the *same* series.

RMSE vs. MAE: RMSE punishes large errors harder. When big misses are especially costly — a stockout, an understaffed shift — prefer RMSE.

MASE is the one to report across series: $\text{MASE} < 1$ beats seasonal naïve, $\text{MASE} > 1$ is **worse** than doing nothing clever. It is the standard in the M-Competitions (Makridakis et al. 2020).

Scale-free:

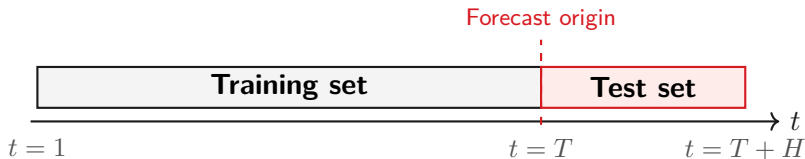
$$\text{MAPE} = \frac{100\%}{H} \sum_{h=1}^H \left| \frac{e_{T+h}}{y_{T+h}} \right|$$

$$\text{MASE} = \frac{\text{MAE}}{\text{MAE}_{\text{seas. naïve}}}$$

Compare *across* series.

The Golden Rule: Always Evaluate Out-of-Sample

Never evaluate a forecast model on the data used to fit it. In-sample fit measures how well the model memorized the past, not how well it predicts the future.



Never randomly shuffle time series observations. Use `TimeSeriesSplit` in `scikit-learn`, never `KFold`. Shuffling lets future data train the model — data leakage that makes results look excellent and forecast badly.

Course Roadmap

What we will cover, and how it fits together.

ECON 8310: Four-Part Structure

Part	Lectures
I. Time series	1. Intro, ETS & evaluation ← today 2. ARIMA & VAR 3. Generalized additive models
II. Tree-based	4. Decision trees 5. Tree ensembles: random forests & boosted trees 6. Regularization & model selection
III. Deep learning	7. Neural networks (PyTorch) 8. CNN architectures 9. RNNs, LSTMs & Transformers
IV. Bayesian	10. Foundations 11. Time series & hierarchical models 12. Bayesian linear regression

Python throughout: statsmodels, scikit-learn, xgboost, PyTorch, prophet, pyGAM, pymc.

Tools and What You Will Be Able to Do

The Python stack: statsmodels (ETS, ARIMA, VAR) · scikit-learn (models, pipelines, cross-validation) · xgboost (gradient boosting) · torch (neural networks) · prophet and pygam (GAMs) · pymc (Bayesian models).

By the end of the semester you will be able to:

- ✓ Choose the right method for a given business problem
- ✓ Implement and tune it in Python
- ✓ Evaluate it rigorously out-of-sample
- ✓ Interpret the result for a non-technical stakeholder
- ✓ Use AI tools effectively to accelerate the analysis
- ✓ Avoid the most common forecasting mistakes

Lecture 1: Key Takeaways

1. Forecasting underpins every forward-looking business decision. The optimal point forecast (under squared loss) is the **conditional expectation**.
2. **Always beat a benchmark first.** Naïve, seasonal naïve, mean, drift.
3. Exponential smoothing assigns **geometrically declining weights** to past observations. α controls how fast old data is forgotten.
4. SES (level only) $\rightarrow$ Holt (+ trend) $\rightarrow$ Holt-Winters (+ season). The **ETS framework** unifies all variants; AIC selects automatically.
5. **Evaluate out-of-sample only.** RMSE penalizes large errors; MASE benchmarks against seasonal naïve. Never shuffle time series data.

Next: ARIMA & VAR — a regression-based approach to time series with rich autocorrelation structure.

References I

-  **Holt, Charles C. (2004).** “Forecasting Seasonals and Trends by Exponentially Weighted Moving Averages”. In: *International Journal of Forecasting* 20.1. Reprint of the 1957 ONR Research Memorandum, pp. 5–10.
-  **Hyndman, Rob J. and George Athanasopoulos (2021).** *Forecasting: Principles and Practice*. 3rd. Open access. Melbourne, Australia: OTexts. URL: <https://otexts.com/fpp3/>.
-  **Makridakis, Spyros, Evangelos Spiliotis, and Vassilios Assimakopoulos (2020).** “The M4 Competition: 100,000 Time Series and 61 Forecasting Methods”. In: *International Journal of Forecasting* 36.1, pp. 54–74.
-  **Winters, Peter R. (1960).** “Forecasting Sales by Exponentially Weighted Moving Averages”. In: *Management Science* 6.3, pp. 324–342.